Apply filters to SQL queries

Project description

In this project, I use SQL queries with filtering conditions to retrieve specific information from database tables. I apply WHERE, AND, OR, and NOT operators to narrow down results based on given criteria such as time, location, and department.

First, you'll retrieve all failed login attempts after business hours. **Second**, you'll retrieve all login attempts that occurred on specific dates. **Third**, you'll retrieve logins that didn't originate in Mexico. **Fourth**, you'll retrieve information about certain employees in the Marketing department. **Fifth**, you'll retrieve information about employees in the Finance or the Sales department. **Finally**, you'll obtain information about employees who are not in the Information Technology department.

Database Structure

The database contains two tables: log_in_attempts and employees.

log_in_attempts

The log_in_attempts table has the following columns:

- event_id: The identification number assigned to each login event
- username: The username of the employee
- login_date: The date the login attempt was recorded
- login_time: The time the login attempt was recorded
- country: The country where the login attempt occurred
- ip_address: The IP address of that employee's machine
- success: The success of the login attempt; FALSE indicates a failed attempt

In the MariaDB shell, these columns are returned as:

```
+----------+----------+------------+------------+---------+-----------------+---------+
| event_id | username | login_date | login_time | country | ip_address      | success |
+----------+----------+------------+------------+---------+-----------------+---------+
```

employees

The employees table has the following columns:

- employee_id: The identification number assigned to each employee
- device_id: The identification number assigned to each device used by the employee
- username: The username of the employee
- department: The department the employee is in
- office: The office the employee is located in

In the MariaDB shell, these columns are returned as:

```
+-------------+-------------+----------+----------------------+-------------+
| employee_id | device_id   | username | department           | office      |
+-------------+-------------+----------+----------------------+-------------+
```

Retrieve after-hours failed login attempts

The security team investigates failed login attempts occurring after business hours (after 18:00).
To investigate this, I wrote a query:

```
MariaDB [organization]> clear
MariaDB [organization]> SELECT *
    -> FROM log_in_attempts
    -> WHERE login_time > '18:00' AND success = FALSE;
```

Use the **AND** operator to retrieve the failed login attempts that occurred after business hours. There are 19 failed login attempts that occurred after 18:00.

```
MariaDB [organization]> SELECT *
    -> FROM log_in_attempts
    -> WHERE login_time > '18:00' AND success = FALSE;
+----------+----------+------------+------------+---------+-----------------+---------+
| event_id | username | login_date | login_time | country | ip_address      | success |
+----------+----------+------------+------------+---------+-----------------+---------+
|        2 | apatel   | 2022-05-10 | 20:27:27   | CAN     | 192.168.205.12  |       0 |
|       18 | pwashing | 2022-05-11 | 19:28:50   | US      | 192.168.66.142  |       0 |
|       20 | tshah    | 2022-05-12 | 18:56:36   | MEXICO  | 192.168.109.50  |       0 |
|       28 | aestrada | 2022-05-09 | 19:28:12   | MEXICO  | 192.168.27.57   |       0 |
|       34 | drosas   | 2022-05-11 | 21:02:04   | US      | 192.168.45.93   |       0 |
|       42 | cgriffin | 2022-05-09 | 23:04:05   | US      | 192.168.4.157   |       0 |
|       52 | cjackson | 2022-05-10 | 22:07:07   | CAN     | 192.168.58.57   |       0 |
|       69 | wjaffrey | 2022-05-11 | 19:55:15   | USA     | 192.168.100.17  |       0 |
|       82 | abernard | 2022-05-12 | 23:38:46   | MEX     | 192.168.234.49  |       0 |
|       87 | apatel   | 2022-05-08 | 22:38:31   | CANADA  | 192.168.132.153 |       0 |
|       96 | ivelasco | 2022-05-09 | 22:36:36   | CAN     | 192.168.84.194  |       0 |
|      104 | asundara | 2022-05-11 | 18:38:07   | US      | 192.168.96.200  |       0 |
|      107 | bisles   | 2022-05-12 | 20:25:57   | USA     | 192.168.116.187 |       0 |
|      111 | aestrada | 2022-05-10 | 22:00:26   | MEXICO  | 192.168.76.27   |       0 |
|      127 | abellmas | 2022-05-09 | 21:20:51   | CANADA  | 192.168.70.122  |       0 |
|      131 | bisles   | 2022-05-09 | 20:03:55   | US      | 192.168.113.171 |       0 |
|      155 | cgriffin | 2022-05-12 | 22:18:42   | USA     | 192.168.236.176 |       0 |
|      160 | jclark   | 2022-05-10 | 20:49:00   | CANADA  | 192.168.214.49  |       0 |
|      199 | yappiah  | 2022-05-11 | 19:34:48   | MEXICO  | 192.168.44.232  |       0 |
+----------+----------+------------+------------+---------+-----------------+---------+
19 rows in set (0.001 sec)

MariaDB [organization]>
```

Retrieve login attempts on specific dates

The team is investigating a suspicious event that occurred on '2022-05-09'. You want to retrieve all login attempts that occurred on this day and the day before ('2022-05-08').

```
MariaDB [organization]> SELECT *
    -> FROM log_in_attempts
    -> WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
```

The **OR** operator return both days.

```
|      170 | sbaelish | 2022-05-09 | 16:43:18   | USA     | 192.168.65.113  |       0 |
|      172 | mabadi   | 2022-05-08 | 08:06:50   | US      | 192.168.180.41  |       1 |
|      178 | sgilmore | 2022-05-08 | 12:27:22   | CAN     | 192.168.52.216  |       0 |
|      184 | alevitsk | 2022-05-08 | 03:09:48   | CAN     | 192.168.33.70   |       0 |
|      186 | bisles   | 2022-05-09 | 04:29:17   | USA     | 192.168.40.72   |       0 |
|      187 | arusso   | 2022-05-09 | 00:36:26   | MEX     | 192.168.77.137  |       0 |
|      189 | nmason   | 2022-05-08 | 05:37:24   | CANADA  | 192.168.168.117 |       1 |
|      190 | jsoto    | 2022-05-09 | 05:09:21   | USA     | 192.168.25.60   |       0 |
|      191 | cjackson | 2022-05-08 | 06:46:07   | CANADA  | 192.168.7.187   |       0 |
|      193 | lrodriqu | 2022-05-08 | 07:11:29   | US      | 192.168.125.240 |       0 |
|      197 | jsoto    | 2022-05-08 | 09:05:09   | US      | 192.168.36.21   |       0 |
+----------+----------+------------+------------+---------+-----------------+---------+
75 rows in set (0.001 sec)
```

There are 75 login attempts in these two days.

Retrieve login attempts outside of Mexico

The team is investigating logins that did not originate in Mexico, and running a query using the NOT and LIKE operators. Because the country field includes entries with 'MEX' and 'MEXICO' we are matching the pattern 'MEX%'.

```
MariaDB [organization]> SELECT *
    -> FROM log_in_attempts
    -> WHERE NOT country LIKE 'MEX%';
```

```
|      194 | jclark   | 2022-05-12 | 14:11:04   | CAN     | 192.168.197.247 |       0 |
|      195 | alevitsk | 2022-05-11 | 06:59:13   | CANADA  | 192.168.236.78  |       1 |
|      196 | acook    | 2022-05-10 | 09:56:48   | CAN     | 192.168.52.90   |       0 |
|      197 | jsoto    | 2022-05-08 | 09:05:09   | US      | 192.168.36.21   |       0 |
|      200 | jclark   | 2022-05-12 | 01:11:45   | CANADA  | 192.168.91.103  |       1 |
+----------+----------+------------+------------+---------+-----------------+---------+
144 rows in set (0.049 sec)
```

There are 144 login attempts made outside of Mexico.

Retrieve employees in Marketing

Retrieve employees in Marketing

The team is updating employee machines, and you need to obtain the information about employees in the 'Marketing' department who are located in all offices in the East building (such as 'East-170' or 'East-320').

I need to use the AND and LIKE operators to satisfy both of these criteria.

```
MariaDB [organization]> SELECT *
    -> FROM employees
    -> WHERE department = 'Marketing' AND office LIKE 'East%';
+-------------+---------------+----------+------------+----------+
| employee_id | device_id     | username | department | office   |
+-------------+---------------+----------+------------+----------+
|        1000 | a320b137c219  | elarson  | Marketing  | East-170 |
|        1052 | a192b174c940  | jdarosa  | Marketing  | East-195 |
|        1075 | x573y883z772  | fbautist | Marketing  | East-267 |
|        1088 | k8651965m233  | rgosh    | Marketing  | East-157 |
|        1103 | NULL          | randerss | Marketing  | East-460 |
|        1156 | a184b775c707  | dellery  | Marketing  | East-417 |
|        1163 | h679i515j339  | cwilliam | Marketing  | East-216 |
+-------------+---------------+----------+------------+----------+
7 rows in set (0.001 sec)
```

Retrieve employees in Finance or Sales

Now, your team needs to perform a different update to the computers of all employees in the Finance or the Sales department, and you need to locate information on these employees.

> To write a SQL query to retrieve records for employees in the 'Finance' or the 'Sales' department, *though both conditions are based on the same column, we need to write out both full conditions.*

```
MariaDB [organization]> SELECT *
    -> FROM employees
    -> WHERE department = 'Finance' OR department = 'Sales';
```

```
MariaDB [organization]> SELECT *
    -> FROM employees
    -> WHERE department = 'Finance' OR department = 'Sales';
+-------------+---------------+----------+------------+-------------+
| employee_id | device_id     | username | department | office      |
+-------------+---------------+----------+------------+-------------+
|        1003 | d394e816f943  | sgilmore | Finance    | South-153   |
|        1007 | h174i497j413  | wjaffrey | Finance    | North-406   |
|        1008 | i858j583k571  | abernard | Finance    | South-170   |
|        1009 | NULL          | lrodriqu | Sales      | South-134   |
|        1010 | k2421212m542  | jlansky  | Finance    | South-109   |
|        1011 | 1748m120n401  | drosas   | Sales      | South-292   |
|        1015 | p611q262r945  | jsoto    | Finance    | North-271   |
```

```
|        1185 | d790e839f461  | revens   | Sales      | North-330   |
|        1186 | e281f433g404  | sacosta  | Sales      | North-460   |
|        1187 | f963g637h851  | bbode    | Finance    | East-351    |
|        1188 | g164h566i795  | noshiro  | Finance    | West-252    |
|        1195 | n516o853p957  | orainier | Finance    | East-346    |
+-------------+---------------+----------+------------+-------------+
71 rows in set (0.001 sec)
```

Retrieve all employees not in IT

The team needs to make one more update. This update was already made to employee computers in the Information Technology department. The team needs information about employees who are not in that department. I use the NOT operator to identify these employees to exclude the Information Technology department.

> a SQL query to retrieve records for employees who are not in the 'Information Technology' department.

```
MariaDB [organization]> SELECT *
    -> FROM employees
    -> WHERE NOT department = 'Information Technology';
+-------------+--------------+----------+-----------------+-------------+
| employee_id | device_id    | username | department      | office      |
+-------------+--------------+----------+-----------------+-------------+
```

Summary

In this project, I practiced using SQL queries to retrieve specific data from a database by applying filtering conditions. I used operators such as WHERE, AND, OR, NOT, and LIKE to refine query results. These techniques are essential for analyzing login activity and managing employee data efficiently.